Hierarchical pooled MLP

Public combat observations → structured encoding → policy and value

EXPERIMENTAL

No content vocabulary

Public inputs · Shared encoders · Learned vectors · Pooling / heads · Retained entities

a Typed observation processing

Player & global context
HP · block · resources · powers
Counters · threat · choice state

Cards, grouped by pile
Current cost · type · modifiers
Public printed effects [E]

Enemies
HP · block · displayed intent
Owned powers · recent history

Potions & relics
Availability · public counters
Public effect descriptions [E]

Context / entity encoders
Separate typed inputs

Shared card encoder
Same weights for every card

Shared enemy encoder
History stays with its enemy

Typed item encoders
Shared within potion / relic type

Globals + entity tokens
Player / selection operands

Card embeddings
Ncards × d

Enemy embeddings
Nenemies × d

Item embeddings
Nitems × d

Keep context separate
Globals · player · selection

Separate pool for each pile
Hand · draw · discard · exhaust
Offered · in-play · sequestered

Enemy pool
Living / present masks retained

Separate item pools
Potions and relics

Concatenate globals and group summaries

Nonlinear state MLP → state vector z

Collection pools use mean + max + log-count. State / entity width d = 64.

b Shared mechanics module [E] — used inside card and item encoders

Effect records
Kind · trigger · target · duration
Amount · hits · expression
Known / applicability flags

Shared effect MLP
Same weights for
every effect record

Ordered combination
Preserve effect sequence
and dependencies

Mechanics vector
Into its owning
card / item encoder

Mechanical categories use fixed one-hot fields; content names and content IDs are excluded.

c Policy and value heads

State vector z
From the state MLP in (a)

Value head

State value V(s)
Both heads see offered choices

Legal candidate table
Action type · subject / target refs

Retained entity embeddings
From (a), including offered choices

Gather candidate operands
Subject / target embeddings + action

same state z for each candidate

Shared candidate scorer

Candidate masks
Legal / policy constraints

Masked softmax → π(a | s)
Distribution over candidate actions

Experimental architecture; see the study record for evidence. Layout inspired by OpenAI Five, Fig. 17.